Andrew Yi Zhou

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EDUCATION

University of California San Diego **Sept. 2023 - June 2027**

B.S. in Computer Science, B.S. in Applied Mathematics, Minor in Music

- **GPA**: 3.98
- **Coursework**: Deep Learning, Reinforcement Learning, NLP, Computer Architecture, Operating Systems, Statistics

PUBLICATIONS

- **Andrew Y. Zhou**, Sharvaree Vadgama, Sumanth Varambally, Peter Eckmann, Michael K. Gilson, Rose Yu , "ToolMol: Evolutionary Agentic Framework for Multi-objective Drug Discovery", *https://arxiv.org/abs/2605.12784*

EXPERIENCE

Spatiotemporal Machine Learning Lab, San Diego, CA **Feb. 2025 - Present**

Machine Learning Research Intern

- Advised by Dr. Rose Yu and Dr. Mike Gilson in developing **LLMs** for application to **small molecule drug discovery**
- Implemented post-training through **Supervised Fine-Tuning**, **Reinforcement Learning (PPO), LoRA, Distillation** and **RAG** to optimize for molecular generation, achieving over **20% affinity improvement** over SOTA methods
- First-authored paper on **agentic genetic algorithm framework** using function-calling and large database retrieval

Scripps Institution of Oceanography, San Diego, CA **Nov. 2024 - Present**

Machine Learning Research Lead

- Leading collaboration with Scripps Institution of Oceanography to develop **deep computer vision segmentation models** to remotely monitor health of **critical carbon-rich mangrove populations** in global areas of interest
- Built **UNet classifier** w/ ResNet18 encoder that achieves >**95% test accuracy** on high resolution drone imagery
- Developed **super-resolution** techniques (**diffusion**, **SRCNN**, **vision transformer**) for lower resolution satellite data

UCSD Center for Energy Research, San Diego, CA **June 2024 - Dec. 2024**

Software Engineer Intern

- Worked with Research Director Michael Ferry to create data visualizations of climate change in upcoming decades
- Lead developer for the **Triton Energy Climate** app, an interactive mobile app on App Store and Google Play that projects the effects of increasing renewable energy on CO_2 emissions; build with **React Native** and **PostgreSQL**
- Presented at November 2024 **United Nations COP29 Climate Change Conference** with **100+ users**

PROJECTS

Veteran Support Website **Jan. 2025 - June 2025**

- Created **full-stack application** for non-profit organization Veteran Sisters to facilitate management of veteran clients
- Built extensive **custom PDF viewer** with commenting support in **React.js** for ease of document access by admins
- Managed all document storage operations with **Firebase**, built robust API routes for controlled **MongoDB** interaction

Housing Portal Application **Jan. 2024 - June 2024**

- Developed **full-stack admin interface** for non-profit organization Union Station Homeless Services to streamline connection process between landlords and homeless tenants, as well as facilitate management of housing units
- Engineered frontend logic (**dynamic filtering**, **non-authenticated redirection**, etc) and **Express.js backend API**
- Implemented **OAuth 2.0** for smooth **user authentication** and remote storage of large photos/videos for housing units

SKILLS & INTERESTS

- Research Interests: **Deep Generative Modeling**, **LLMs**, **Ai4Science, Computational Drug Discovery**
- Programming languages: **Java**, **C, C++**, **Python**, **JavaScript, TypeScript, HTML/CSS, x86 & ARM Assembly**
- Machine learning: **PyTorch**, **Tensorflow**, **CUDA**, **NumPy**, **pandas, OpenCV, vLLM, HuggingFace, Quantization**
- Full-stack development: **React.js, Express.js, MongoDB, Node.js, PostgreSQL, Docker, Kubernetes, AWS**